

Product Carbon Footprint Study Report

Product Name: ■■■■

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: ■■■■

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - IPCC 2021

II. Quantitative Purpose

1 kg Ethanol

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg Ethanol.

2. System Boundary

1 kg Ethanol production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: ■■■■

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of ■■■■■ produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **5.229 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

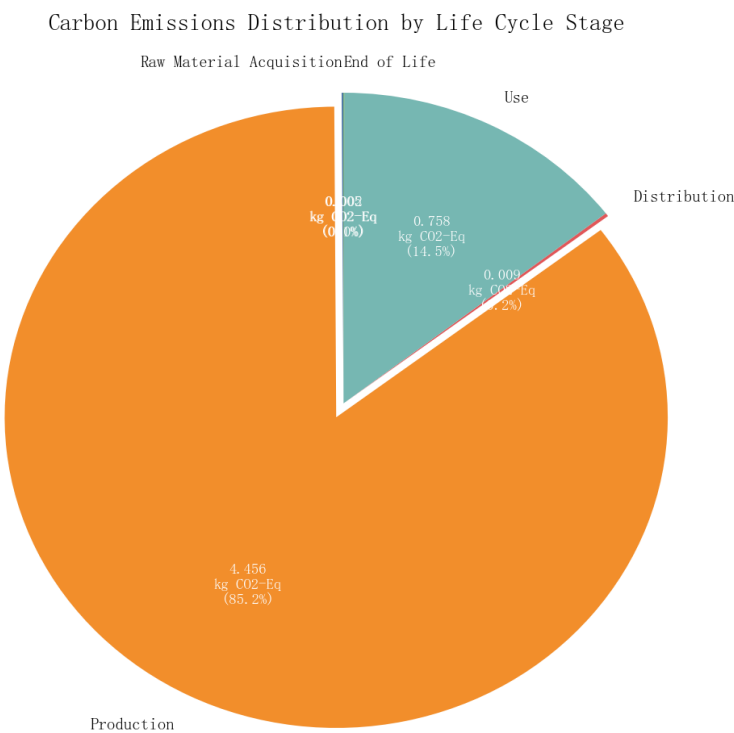


Table 1: Carbon Emissions by Life Cycle Stage

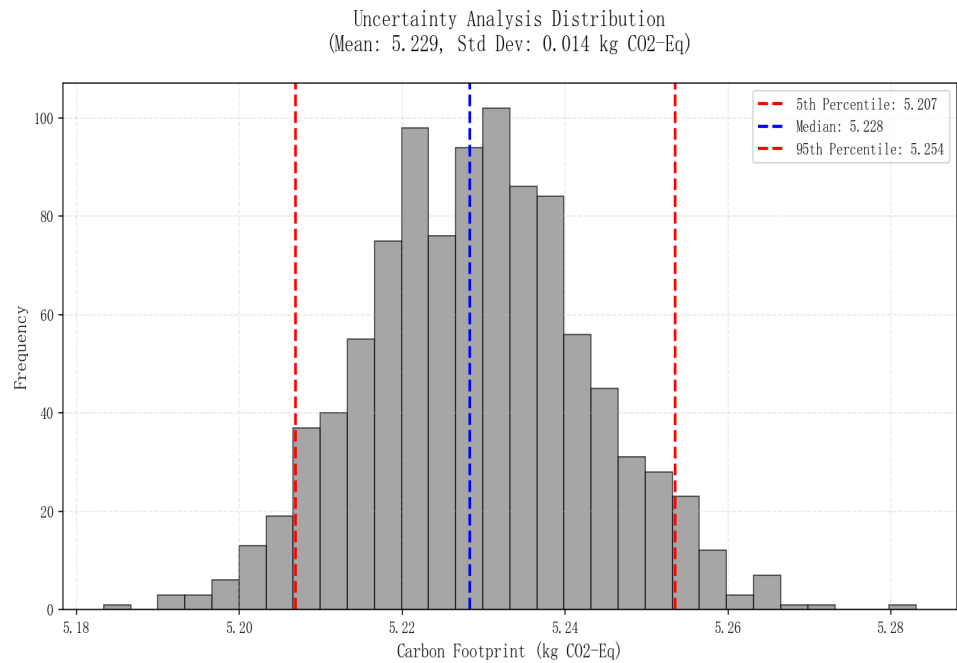
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.005	0.09
Production	4.456	85.21
Distribution	0.009	0.17

Use	0.758	14.49
End of Life	0.002	0.04
Total	5.229	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	5.229
Standard Deviation	0.014
Coefficient of Variation (CV)	0.27%
Median	5.228
5th Percentile	5.207
95th Percentile	5.254

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of ethanol production is 5.229 kg CO₂-Eq, representing a moderate emission intensity for this chemical product under the assessed functional unit. This value reflects a cradle-to-grave evaluation encompassing all major lifecycle phases. The distribution of impacts reveals a highly concentrated profile, with the production stage accounting for 4.456 kg CO₂-Eq or 85.2% of the total, followed by the use stage at 0.758 kg CO₂-Eq (14.5%). Raw material extraction, distribution, and end-of-life phases contribute negligibly, each below 0.2%.

This contribution pattern identifies the production phase as the dominant emission source, likely driven by energy-intensive processes such as fermentation, distillation, and steam generation. The use phase represents secondary impacts, possibly linked to downstream combustion or application emissions. Overall, the results indicate that targeted interventions in manufacturing operations would yield the greatest reduction potential, while upstream and downstream stages offer limited leverage for decarbonization.

2. Statement of Assumptions and Limitations

This assessment assumes average technological performance and standard operating conditions typical of modern ethanol facilities, including conventional energy mixes and process efficiencies. It further presumes that the functional unit corresponds to one kilogram of ethanol output and that allocation methods for co-products follow ISO 14044 guidelines. Data inputs are modeled using generic background inventories rather than site-specific measurements.

Limitations include reliance on secondary databases that may not fully capture regional variations in feedstock sourcing or technological upgrades. Data quality for minor stages is lower due to sparse primary measurements, potentially introducing uncertainty in the negligible contributions reported. Temporal representativeness may also be constrained by older reference datasets.

System boundaries are restricted to a cradle-to-grave scope covering raw material acquisition through end-of-life treatment. Exclusions encompass infrastructure construction, capital goods, and indirect land-use changes, which could alter the overall footprint if included. The analysis does not extend to water consumption or toxicity impacts beyond climate change.

3. Suggestions for Improvement

- Replace fossil-based process heat in the production stage with renewable electricity or biomass boilers to address the 85.2% contribution from manufacturing.
- Implement heat integration and advanced distillation technologies to lower energy demand during ethanol purification.
- Optimize the use phase through efficiency improvements in downstream applications, such as blending protocols that minimize combustion emissions.
- Conduct primary data collection at the production facility to refine emission factors and reduce reliance on generic datasets.

- Expand system boundaries in future studies to include land-use change impacts for more comprehensive feedstock evaluation.

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